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HUJI Applicative Research Report — Exact & Social Sciences — Week 34, 2026

2 paper(s) · Exact & Social Sciences · Weekly · Week 34, 2026 · Generated August 17, 2026

■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This week's digest highlights groundbreaking research from Hebrew University, showcasing innovations with significant commercial promise. We feature advancements in quantum computing, specifically universal Hamiltonian simulators, poised to revolutionize drug discovery and materials science by accelerating high-fidelity simulations. Concurrently, we present critical research in AgriTech identifying a hidden pathway of environmental contamination from tire wear, alongside the promising opportunity for developing targeted remediation technologies. These diverse initiatives underscore our commitment to pioneering solutions across high-tech and essential environmental sectors.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Dorit Aharonov** — Quantum, Materials, Software/AI, Drug Discovery
- **Evyatar Ben Mordechay** — AgriTech, Materials, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Universal Hamiltonian simulators in one and two dimensions. 24/50

"Quantum simulation breakthroughs promise to accelerate drug and materials discovery, unlocking new product development."

Main researcher: Dorit Aharonov

School of Computer Science and Engineering, The Hebrew University, Jerusalem, Israel.

Published in Nature communications · 2026-04

ABOUT THIS RESEARCH

This research presents a way to use quantum computers to simulate complex physical systems by creating universal simulators in one and two dimensions. This allows scientists to model the behavior of quantum particles and energy states much more accurately than classical computers can.

COMMERCIAL ANGLE

The ability to perform high-fidelity analog quantum simulations could drastically accelerate the computational discovery of new molecules and materials by modeling their quantum properties from first principles.

COMMERCIALISATION METRICS

Novelty 8/10

The work introduces a platform-enabling methodology for universal analog simulation across different dimensions, representing a significant conceptual leap over dimension-specific or non-universal prior art. It provides a scalable framework that is highly relevant for industrial quantum advantage in material science and chemistry.

Commercial Potential 4/10

The paper focuses on fundamental theoretical physics and algorithmic frameworks for quantum simulation, which lacks a direct pathway to a specific industrial application or product. While it may enable future quantum software platforms, the commercial utility is highly speculative and dependent on the long-term scaling of quantum hardware.

Market Size 7/10

The work targets the fundamental quantum simulation market, which is a foundational pillar of the multi-billion dollar quantum computing industry. While the potential impact on drug discovery and materials science is massive, the specific focus on Hamiltonian simulators addresses a specialized niche within the broader hardware/software stack.

Tech Readiness 2/10

The paper describes fundamental theoretical physics and algorithmic frameworks for quantum simulation, which is at the earliest stage of mathematical proof rather than hardware implementation. It lacks any empirical validation on physical quantum hardware or proximity to a functional commercial platform.

IP Strength 3/10

The work is fundamental theoretical physics focused on algorithmic frameworks, which are notoriously difficult to patent and often fall under non-patentable mathematical abstractions. While scientifically significant, the innovation lacks the specific hardware implementation or industrial utility required to build a defensible IP moat.

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2. Tire wear particles as a source of organic contaminants in the agro-environment: Release of tire wear-derived compounds and their plant uptake. 15/50

"New agri-tech solutions needed to combat tire microplastic contamination and ensure crop safety."

Main researcher: Evyatar Ben Mordechay · evyatar.mordechay@mail.huji.ac.il.

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Published in Environmental research · 2026 Aug 15

ABOUT THIS RESEARCH

The research investigates how microplastics and organic chemicals from tire wear enter agricultural soils and are subsequently absorbed by plants. It highlights a hidden pathway of environmental contamination that could affect crop safety and soil health.

COMMERCIAL ANGLE

Development of advanced soil remediation technologies or bio-filtration systems designed to sequester tire-derived contaminants from agricultural land.

COMMERCIALISATION METRICS

Novelty 5/10

The research applies established environmental chemistry and plant physiology methods to a specific emerging contaminant (tire wear particles), representing an incremental expansion of known pollutant pathways rather than a groundbreaking scientific discovery.

Commercial Potential 2/10

This is basic environmental toxicology research identifying a pollutant source; it describes a problem rather than providing a protectable solution, product, or scalable technology.

Market Size 4/10

The paper addresses a niche environmental contamination issue; while tire wear is global, the immediate addressable market for a specific technical solution is limited compared to broader agricultural or industrial platforms.

Tech Readiness 2/10

This is fundamental environmental research focused on identifying contaminants and plant uptake mechanisms; it is at the basic laboratory/observational stage with no mention of a scalable product, mitigation technology, or commercial application.

IP Strength 2/10

The paper describes environmental monitoring and mechanistic research on contaminant uptake, which is fundamental scientific discovery and generally not patentable. There is no evidence of a proprietary technology, platform, or novel industrial process that could be defended as intellectual property.

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